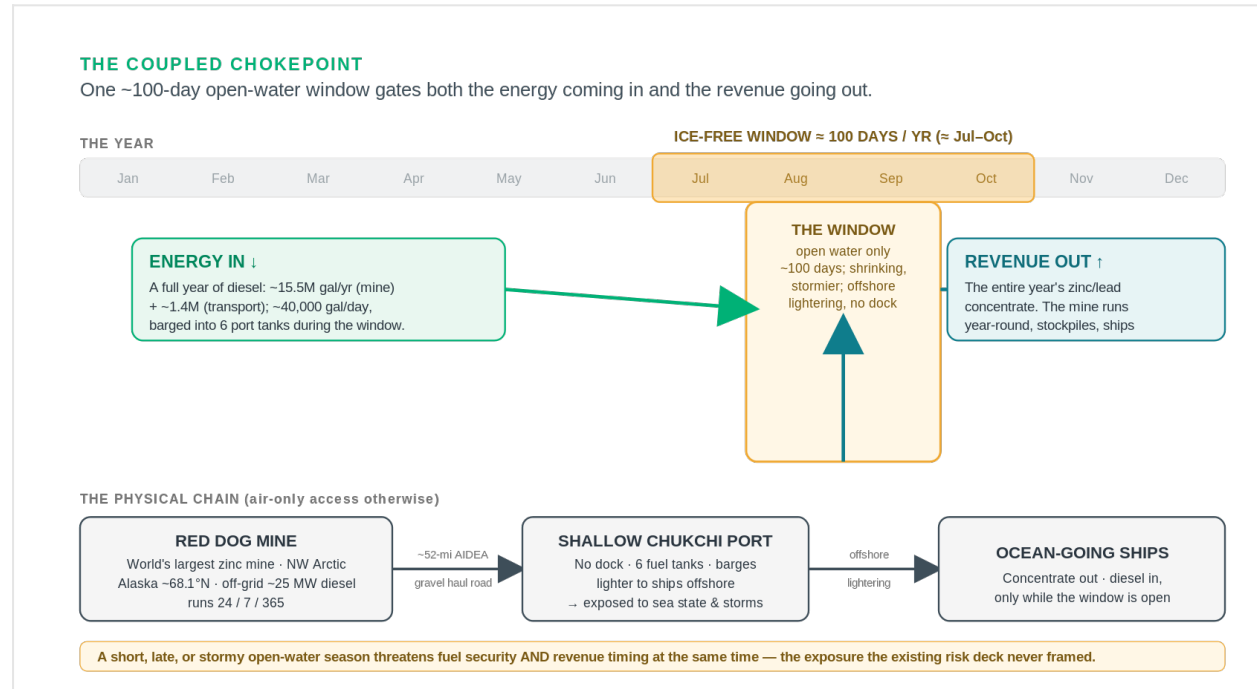


Red Dog Mine — Bottom-Up Climate-Risk Read

A worked, bottom-up climate-risk read on a single off-substrate asset — Teck/NANA's Red Dog Mine, NW Arctic Alaska.

For: InfraSure team (internal) · As of 2026-06-14 · Re-grounded 2026-06-15 · Draft v1 (for review before any external use)



The coupled energy-and-logistics chokepoint — the central thesis in one screen (schematic). Sources: AIDEA; Teck FY2025 production; Alaska DNR. As of 2026-06-15.

TL;DR

Red Dog is a **single-point-of-failure energy-and-logistics asset** in the fastest-warming region on Earth. Everything the mine needs (a year of diesel) and everything it sells (its entire zinc/lead concentrate) moves through **one gravel road to one shallow Chukchi-Sea port that is ice-free only ~100 days a year** — so the same shrinking-and-stormier open-water window gates **both** energy-in and revenue-out. That coupling is the climate exposure the existing risk deck never framed, and it is squarely InfraSure's lane. Layered on top is a **permafrost-driven water-balance crisis** that a US regulator has formally conceded *cannot be fixed* (a 10-year compliance runway to ~2036), pushing a **~\$279M adaptation capex onto a mine with only ~6 years of ore left**, an escalating royalty, and an owner mid-merger into a copper-focused major. The decision-relevant tail risk — a diesel-tank or tailings-dam failure on thawing ground — is exactly the geometry that

destroyed Norilsk in 2020 (~\$2B), and the asset-level numbers that would price it (fuel autonomy, dam factor-of-safety, foundation engineering) are **not disclosed** — a finding in itself.

Re-ground (2026-06-15): the moving facts were re-verified against live public sources. **Updated**: the Anglo–Teck merger (Canada + South Korea now cleared; **only China remains**). **Confirmed**: the ADEC permit (reissuance public-noticed, ~2036 runway) and the forward production decline. **Base check (held the line)**: the company-total 2025 zinc figure is ~565 kt, but that is **Teck-total (Red Dog + Antamina)** — Red Dog's own ~462,700 t below is the right base and is unchanged; the two must not be conflated. **Candidate to verify before external use**: Teck reportedly cites *wetter weather limiting reserve access* among the decline drivers — a direct confirmation of \$3 if matched to Teck's exact wording.

What this is: a worked, *bottom-up* climate-risk read on a single off-substrate asset — Teck/NANA's **Red Dog Mine**, NW Arctic Alaska — produced by the reverse-engineering method (research + the model + our methodology), **not** the InfraSure platform. It is the proof case for handling outliers that fall outside our energy-infrastructure substrate.

Grounding honesty (read first): this is **research-grounded** — public sources (primary regulatory filings, Teck disclosures, peer-reviewed science, reporting) disciplined by our claim/confidence/gate rules. It uses **no** InfraSure MCP data and **no** model-gpr output. The value here is **rigor + currency + the energy-infrastructure framing**, not proprietary data. It deliberately does **not** re-do the existing LT-Risk deck; it adds what the deck lacks.

1 · The asset in one screen

Operator / owner	World's largest zinc mine · Teck Resources operates · NANA Regional Corp. (Iñupiat) owns the land
Location & type	NW Arctic Alaska, ~68.1°N (170 km N of the Arctic Circle) · open-pit zinc/lead · runs 24/7/365 · air-only access
Power	OFF-GRID: ~25 MW typical demand on diesel gensets (installed capacity contested 24.5–40+ MW)

Logistics	~52-mi state-owned (AIDEA) gravel haul road → one shallow Chukchi port (no dock; barges lighter to ships offshore)
Production	FY2025: 462,700 t contained zinc (↓~17% from 555,600 t in 2024, on falling grades) · ore life ~2032

Sources: Alaska DNR; NS Energy; AIDEA; FindAPort; Teck FY2025 production (Feb 2026). Confidence: high (the contested item — generator nameplate — is flagged in §6).

2 · The lead exposure — a coupled energy-and-logistics chokepoint

This is the framing the LT-Risk deck missed, and it's the most decision-relevant climate linkage at the asset. The hero schematic above captures it; in plain terms:

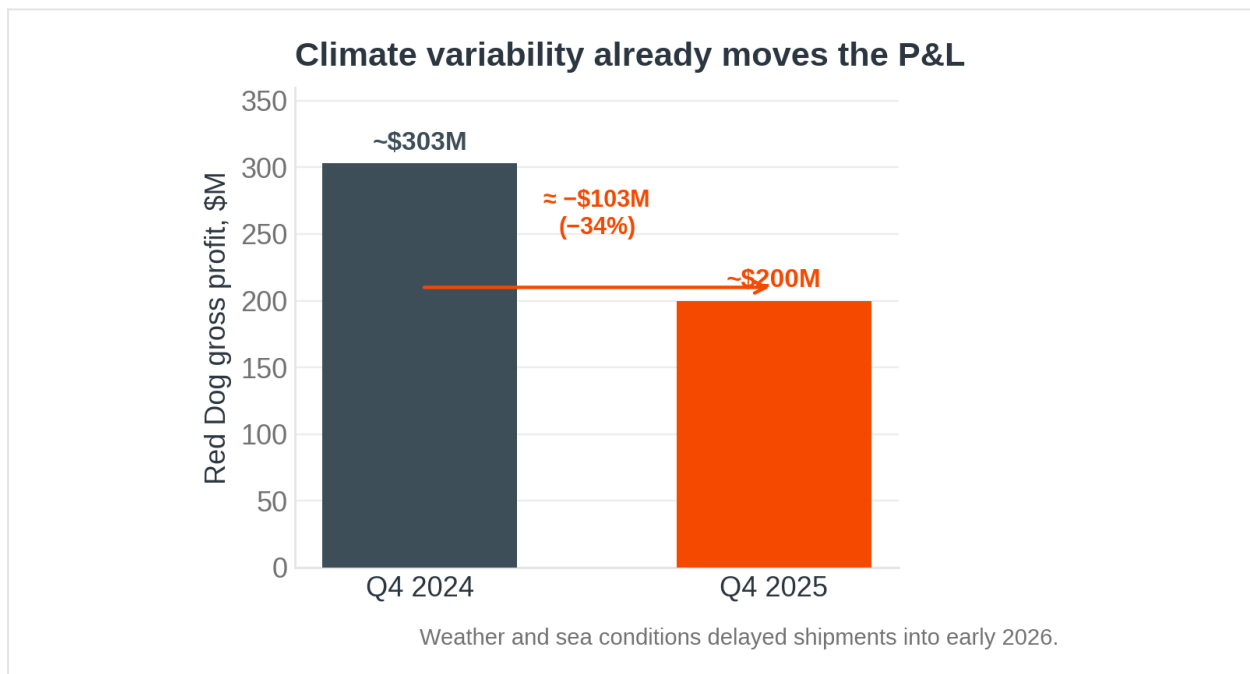
ONE ~100-day ice-free window (≈ July–October) carries BOTH:

Energy-in — the year's diesel: ~15.5M gal/yr (mine) + ~1.4M (transport); ~40,000 gal/day; barged into 6 port tanks.

Revenue-out — the entire annual concentrate export; the mine produces year-round, stockpiles, and ships only in the window.

→ a short, late, or stormy open-water season threatens fuel security AND revenue timing simultaneously.

This is not hypothetical. In late 2025, "weather and sea conditions" delayed shipments into early 2026, and **Q4-2025 Red Dog gross profit fell to ~\$200M (from ~\$303M a year earlier)** on the timing — climate variability already moving the P&L through the logistics window. (Teck Q4-2025 results, Feb 2026 — fact/high.)



Q4 Red Dog gross profit, 2024 vs 2025. Source: Teck Q4-2025 results (Feb 2026). As of 2026-06-15. fact/high.

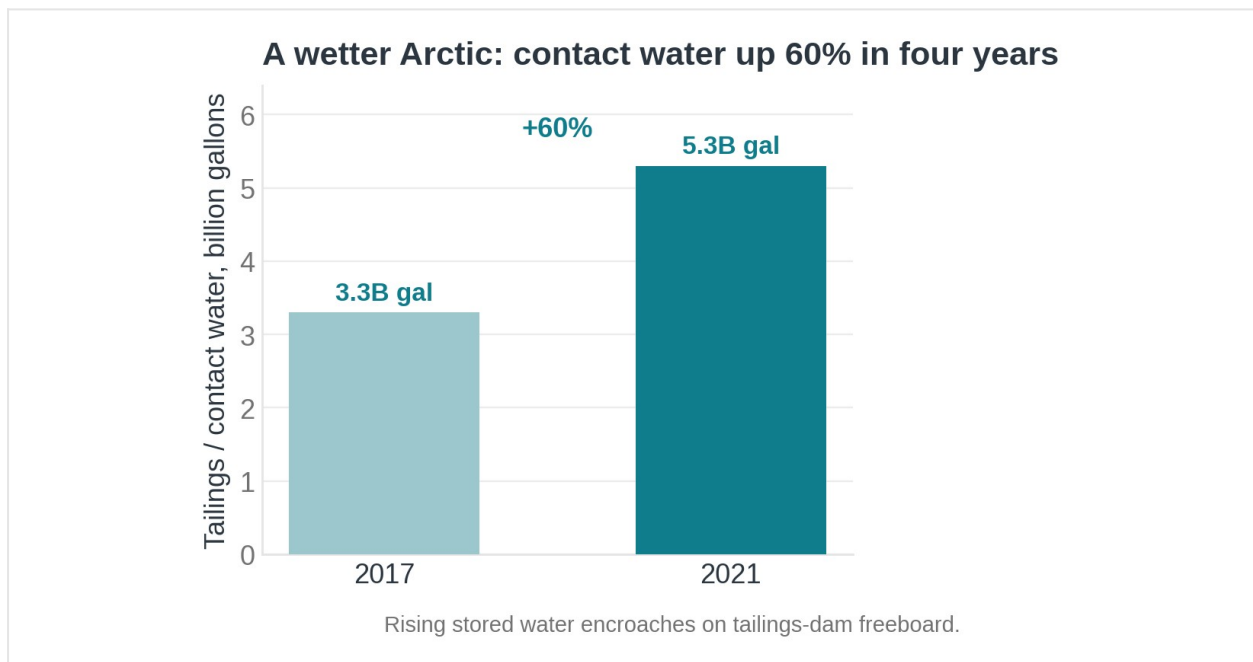
Two structural amplifiers:

- **Sea-ice retreat cuts both ways:** it can lengthen the shipping window (operational upside) *but* removes the landfast-ice buffer that protected the coast — the same coastal cell is eroding fast at nearby Kivalina. The port relies on **offshore lightering**, so it is exposed to *sea state and storms*, not just ice. (*AIDEA; Ground Truth Alaska — fact/medium.*)
- **The single-window dependency is undisclosed at the point that matters:** how many days of fuel the site can run if a resupply season is missed or cut short is **not public** (see §6). An 8.8 MW solar array (Tugliq, ~2 MW summer-effective, tax-credit-timed to start mid-2026) dents diesel but does not change the resupply dependency. (*Alaska Beacon, Mar 2026 — fact/high; the "dents not displaces" read is ours.*)

3 • Permafrost → a water-balance crisis the regulator calls unfixable

The second pillar is documented in **primary regulatory text**, not just press:

permafrost thaw → naturally elevated TDS in receiving waters → lost mixing-zone capacity
 → **discharge blocked**
 + a wetter Arctic → tailings/contact water **+60% (3.3B → 5.3B gallons, 2017–2021)** →
 encroaches on dam freeboard



Tailings / contact water stored, 2017 vs 2021. Source: Alaska DEC APDES AK0038652 Statement of Basis (2026-04-09). As of 2026-06-15. fact/high.

- Alaska DEC is **reissuing the discharge permit (APDES AK0038652) with a 10-year compliance-schedule extension** (public-noticed; effective after May 19, 2026, to ~2036), stating in its own Statement of Basis (2026-04-09) that *"there is no reasonably available remedy to the thawing of permafrost and its impact on the receiving waters."* A US regulator formally conceding a Clean-Water-Act limit can't be met because of permafrost is **decision-relevant and unusual**. (ADEC SOB — fact/high.) **Note:** this corrects the deck's "permit expiring 2026" — it was *extended*, and the runway now *outlives the ~2032 ore life*, shifting the binding constraint toward closure-phase water treatment.
- ADEC states stored water *"threatens the safety of the tailings impoundment dams,"* with a tail risk of *"dam failure"* / untreated-drainage release. The **Aug-2012 storm (15.2 in rain) exceeded the tailings dam's design flood** — the infrastructure has already operated outside its design envelope. (ADEC SOB — fact/high; *this is the single strongest data point that the asset is past-envelope.*)
- The mitigation under evaluation is a **~\$279M, ~50-mile pipeline to the Chukchi Sea** — *"more than half of Red Dog's annual gross profit"* — one of four options, **not committed**. ~\$19M was already spent on thaw-driven wastewater in 2020 alone. (Northern Journal Dec 2025; Alaska Public Media 2020; ADEC SOB — fact/high; *the pipeline is an estimate under evaluation — projection.*)
- The regional driver is real and independent of the mine: **"river rusting"** — thaw-exposed sulfide weathering turning 75+ (200+) Brooks Range streams orange — with on-site seeps

measured at **pH 3.2 and TDS to 11,300 mg/L**. (*O'Donnell 2024 / NOAA Arctic Report Card 2025; ADEC SOB — fact/high.*)

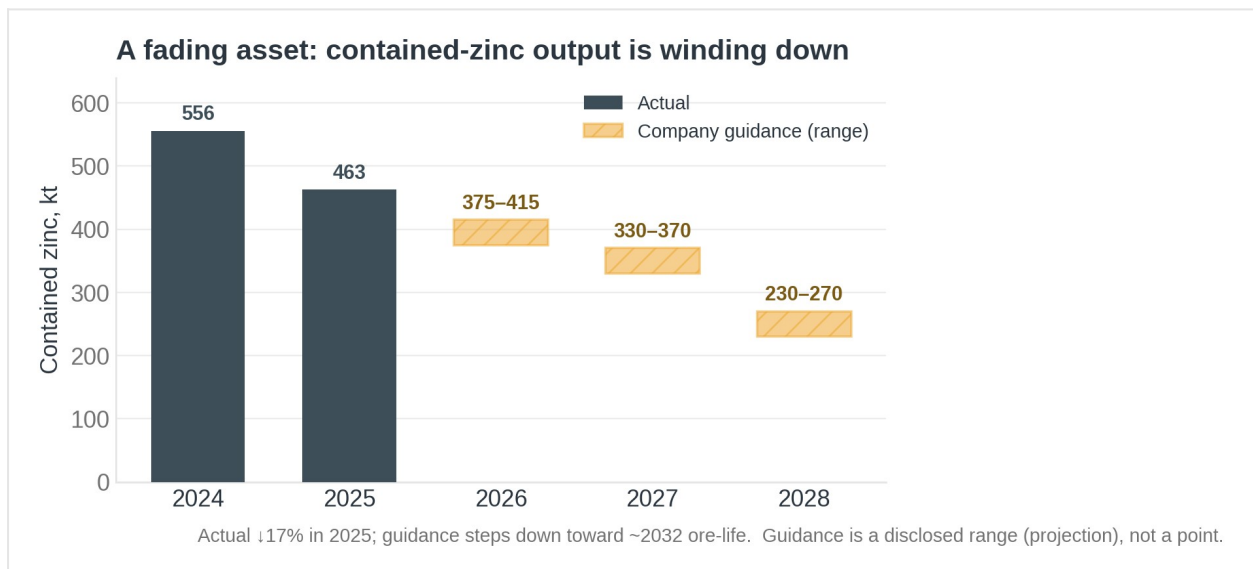
Context for the *rate*: the Arctic has warmed **~4× the global average** since 1979, and climate models *under-state* it — so the signal arrives faster than scenario planning assumes, compressing adaptation lead-time. (*Rantanen 2022, Nature Comms — fact/high.*)

4 • The decision frame — a fading asset asked to fund a costly adaptation

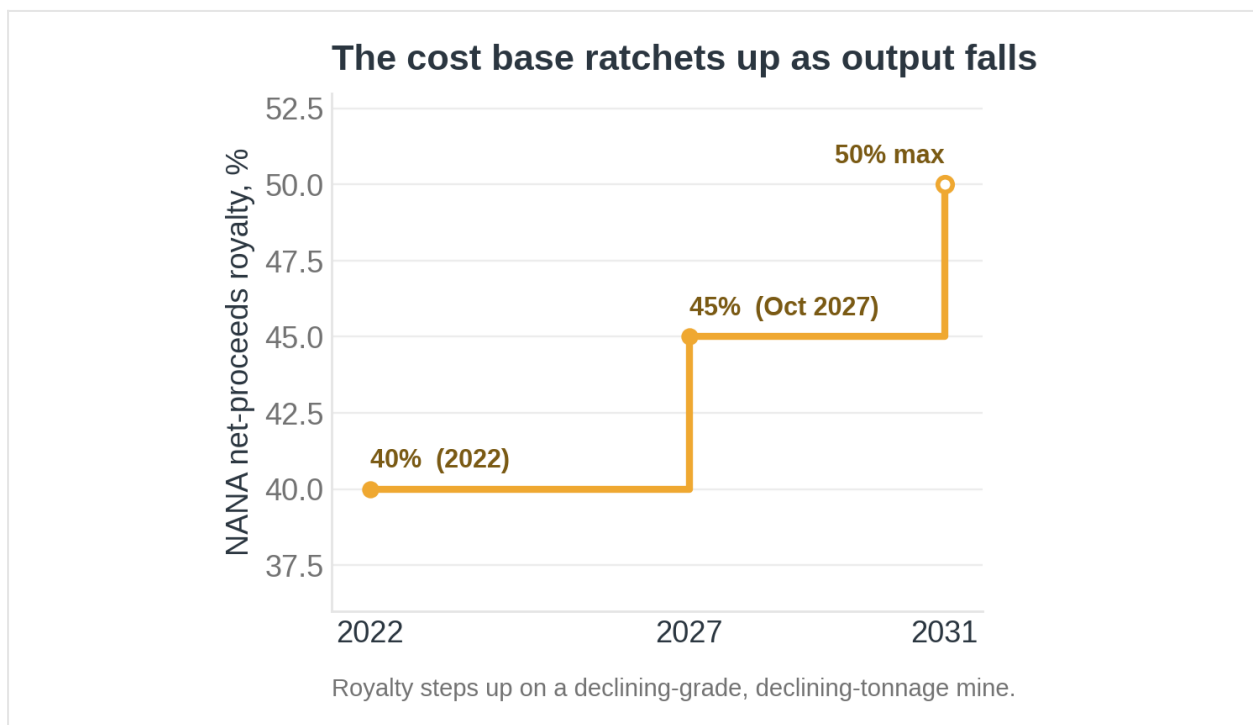
What makes this a genuine decision (not just a hazard list) is the collision of climate capex with late-life economics:

A fading asset	ore ~2032; Red Dog zinc guided DOWN 2026 375–415 kt → 2027 330–370 → 2028 230–270 kt; extension (Aktigiruuq/Anarraaq) only pre-feasibility
A rising cost base	NANA royalty ratchets 40% (2022) → 45% (Oct 2027) → 50% max, as grades/tonnage fall
Ownership in flux	Anglo–Teck merger announced Sept 2025, shareholder-approved Dec 2025; Canada + S. Korea cleared, ONLY CHINA remains (as of Jun 2026), close ~Q3-2026–Q1-2027 → Red Dog a non-core zinc slice of a >70%-COPPER major
A long-tail liability	state closure assurance ~\$558M + PERPETUAL post-closure water treatment (outlives the revenue)

Sources: Teck production guidance + MD&A; North of 60 Mining News; Teck/Anglo merger releases; Alaska DNR. Confidence: high (figures), with mine-life ~2032 and the extension as projections.



Contained-zinc output: actual (2024–25) then company guidance (2026–28). Sources: Teck FY2025 production (Feb 2026); production guidance + MD&A. As of 2026-06-15. Guidance is a disclosed range (projection).



NANA net-proceeds royalty schedule. Sources: Alaska DNR; North of 60 Mining News. As of 2026-06-15. fact/high (2031 step = 50% cap).

The sharp frame, from the Arctic-infrastructure literature: **near-term permafrost risk is largely mitigation-insensitive** — for an *existing* asset, **adaptation, not decarbonization, is the only lever** (Hjort et al. 2018, *Nature Comms*). So the question Red Dog's owner actually faces is **how much adaptation capex to sink into a depleting, soon-to-change-hands asset** — a ~\$279M pipeline (and "hundreds of millions over a decade" for a permanent fix) against ~6

years of declining ore. That tension — not a generic risk rating — is the read a decision-maker needs.

5 · Why this generalizes — the outlier class + a realized precedent

Red Dog instantiates a class InfraSure will meet again: the **remote, off-grid, cold-region industrial asset** — *islanded on a diesel microgrid + dependent on a single climate-sensitive logistics window + built on degrading permafrost*. The method transfers to the next one (another mine, a port, a remote data center).

- **The realized precedent — Norilsk, 2020:** a diesel storage tank on thaw-weakened permafrost piles collapsed → **~21,000 t spill, ~\$2B in damages** — the exact Red Dog risk geometry, realized. (*ERM / press — fact/high.*) This is why the **fuel-farm foundation engineering** (§6) is the most important number we don't have.
- **The mirror precedent — winter ice roads** (Tibbitt–Contwoyto, serving NWT diamond mines): the same narrowing-seasonal-window risk, *opposite sign* (they need ice *present*; Red Dog needs it *absent*) — showing the "single climate-sensitive window" is the portable risk concept. (*Reuters — fact/high.*)

Frameworks that anchor the class: **IPCC AR6 WG2 CCP6** ("warming and wetting"; most Arctic infrastructure impacted by mid-century, high confidence); **Hjort 2018** (33–69% of pan-Arctic infrastructure on thawing permafrost by 2050); **AMAP/SWIPA**.

6 · What we can't know — and why that's a finding

A disciplined read names its blind spots rather than guessing. For Red Dog, the **most decision-relevant numbers are not public**:

Days of fuel autonomy	if an open-water window is missed/short — undisclosed (the latent single-point-of-failure)
Tailings-dam factor-of-safety	current freeboard margin — undisclosed (ADEC says it's threatened; the margin isn't public)
Fuel-farm foundation engineering	piles to bedrock? thermosyphons? — undisclosed (this is exactly the Norilsk failure mode)
Asset-level financials	Red Dog has no standalone public P&L (reported inside Teck's zinc segment; ~CAD \$846M of the segment's \$1,143M FY2025 gross-profit-before-D&A is the closest anchor) — so margin-at-

risk can't be priced precisely

These gaps are the honest boundary of a research-grounded read — and the precise questions a pilot/data-exchange would close.

7 · What InfraSure would actually add here

Honest version, for the team: a generic consultant or a vanilla LLM can produce a Red Dog climate narrative. What this read demonstrates that they don't is **(1) the energy-infrastructure framing** (the coupled diesel-in/revenue-out chokepoint — our lane, the deck's blind spot), **(2) discipline** (every claim sourced, dated, and confidence-rated; the deck's stale points corrected; nothing over-quantified), and **(3) gap-honesty** (we name the undisclosed numbers a pilot would resolve). The platform/MCP and model-gpr do **not** reach this asset — and we say so. That combination *is* the capability worth showing Teck: not "we already have your data," but "here is how rigorously and quickly we reason about an asset we've never been given data for."

Sources & provenance

Primary / regulatory	Alaska DEC APDES AK0038652 Statement of Basis (2026-04-09) · Alaska DNR Large Mines (Red Dog) · AIDEA (DMTS)
Company	Teck Q4/FY2025 results (Feb 2026) · Teck production guidance + MD&A · Teck–Anglo merger releases (Sept/Dec 2025)
Peer-reviewed	Rantanen 2022 (Arctic ~4× warming) · Hjort 2018 (pan-Arctic infra exposure) · O'Donnell 2024 (river rusting) · IPCC AR6 WG2 CCP6
Reporting	Northern Journal / Alaska Beacon / Anchorage Daily News (permafrost, solar, fuel) · Fastmarkets · North of 60 Mining News · Ground Truth Alaska (critical-stakeholder)
Precedent	Norilsk 2020 diesel-tank collapse (ERM / press) · Tibbitt–Contwoyto winter ice road (Reuters)
Re-ground 2026-06-15	Teck 2025 Production & Sales Update (teck.com, 2026-02) · merger: Canada + S. Korea cleared, China pending (MLex; mining.com, 2026) · ADEC AK0038652 reissuance public notice + SOB (2026-04-09)

*Full per-claim provenance (named source · URL · date · confidence · fact-vs-projection) is in the research record behind `docs/discussion/2026-06-14\_outlier\_method\_red\_dog.md`. **Research-grounded; not InfraSure MCP / model-gpr data.***